Continuing from N2 (Kitaev Honeycomb main note), here are a few (algebraic) justifications →

**Hamiltonian:** We have the itinerant $c$-Majorana Hamiltonian:

$$H = \frac{i}{4}\sum_{j,k} A_{jk} c_j c_k$$

$A$ is: Real, AntiSymmetric ($A^T = -A$), of dimension $2N \times 2N$.

**Marginal Note: Why is $A$ antisymmetric?**

Remember that $A_{jk} = ib_j^\alpha b_k^\alpha$ (with $\alpha$ being dependent on the $j-k$ bond type). Since the $b$-Majoranas on different sites anticommute, $\{b_j^\alpha, b_k^\alpha\} = 0$ for $j \neq k$:

$$\begin{aligned} A_{jk} &= ib_j^\alpha b_k^\alpha \\ &= -ib_k^\alpha b_j^\alpha \\ &= -A_{kj} \end{aligned}$$

Also, we can directly demand this as otherwise the Hamiltonian will not be Hermitian.

Let us continue. Define a new Hermitian matrix, $M = iA$. Note:

$$M^\dagger = (iA)^\dagger = -iA^T = iA = M$$

$\implies$ eigenvalues of $M$ are real.

Let us define some eigenvector $|\Psi_m\rangle \in \mathbb{C}^{2N}$, i.e. a state in the extended Hilbert space, that is an eigenvector of $M$:

$$\begin{aligned} M|\Psi_m\rangle &= \epsilon_m|\Psi_m\rangle \\ \implies iA|\Psi_m\rangle &= \epsilon_m|\Psi_m\rangle \\ \implies A|\Psi_m\rangle &= -i\epsilon_m|\Psi_m\rangle \end{aligned}$$

Take complex conjugate on both sides:

$$\begin{aligned} (A|\Psi_m\rangle)^* &= (-i\epsilon_m|\Psi_m\rangle)^* \\ \implies A^*|\Psi_m^*\rangle &= i\epsilon_m|\Psi_m^*\rangle \end{aligned}$$

Since $A$ is real ($A^* = A$), this becomes:

$$\begin{aligned} A|\Psi_m^*\rangle &= i\epsilon_m|\Psi_m^*\rangle \\ \implies iA|\Psi_m^*\rangle &= -\epsilon_m|\Psi_m^*\rangle \\ \implies M|\Psi_m^*\rangle &= -\epsilon_m|\Psi_m^*\rangle \end{aligned}$$

So, the eigenvalues of $M$ arrive in $\pm\epsilon_m$ pairs.

Now, we decompose the $|\Psi_m\rangle$ vector into two real vectors $|u_m\rangle, |v_m\rangle \in \mathbb{R}^{2N}$:

$$|\Psi_m\rangle = |u_m\rangle + i|v_m\rangle$$

Put this into the eigenvalue equation for $|\Psi_m\rangle$:

$$\begin{aligned} iA(|u_m\rangle + i|v_m\rangle) &= \epsilon_m(|u_m\rangle + i|v_m\rangle) \\ \implies iA|u_m\rangle - A|v_m\rangle &= \epsilon_m|u_m\rangle + i\epsilon_m|v_m\rangle \end{aligned}$$

Rearranging to separate real and imaginary components:

$$-A|v_m\rangle + i(A|u_m\rangle) = \epsilon_m|u_m\rangle + i(\epsilon_m|v_m\rangle)$$

Comparing real and imaginary parts:

$$A|u_m\rangle = \epsilon_m|v_m\rangle$$
$$A|v_m\rangle = -\epsilon_m|u_m\rangle$$

Clearly the matrix $A$ acts like a rotation operation on the space spanned by $\{|u_m\rangle, |v_m\rangle\}$.

Assume that $|\Psi_m\rangle$ is normalized, i.e., $\langle\Psi_m|\Psi_m\rangle = 1$:

$$\begin{aligned}\implies \langle\Psi_m|\Psi_m\rangle &= (\langle u_m| - i\langle v_m|)(|u_m\rangle + i|v_m\rangle) \\ &= \langle u_m|u_m\rangle + i\langle u_m|v_m\rangle - i\langle v_m|u_m\rangle + \langle v_m|v_m\rangle = 1 \\ \implies 1 &= [\langle u_m|u_m\rangle + \langle v_m|v_m\rangle] + i[\langle u_m|v_m\rangle - \langle v_m|u_m\rangle]\end{aligned}$$

Since the imaginary part must vanish,

$$\langle u_m|v_m\rangle = \langle v_m|u_m\rangle$$

**Proving orthogonality →**
Take the action of $A$ on $|u_m\rangle$:

$$A|u_m\rangle = \epsilon_m|v_m\rangle$$
$$\langle u_m|A|u_m\rangle = \epsilon_m\langle u_m|v_m\rangle$$

Since $A$ is anti-symmetric, all diagonal elements must vanish $\implies$ LHS is zero.

$$\implies \epsilon_m\langle u_m|v_m\rangle = 0 \implies \langle u_m|v_m\rangle = 0 \quad (\text{As } \epsilon_m \neq 0)$$

**Proving that the norms of $|u_m\rangle$ and $|v_m\rangle$ are equal →**
Use the same equation for the action of $A$ on $|u_m\rangle$:

$$A|u_m\rangle = \epsilon_m|v_m\rangle$$
$$\implies \langle v_m|A|u_m\rangle = \epsilon_m\langle v_m|v_m\rangle$$

Expanding the LHS using the antisymmetry of $A$ ($A^T = -A$) and reality of the vectors:

$$\langle v_m|A|u_m\rangle = \langle u_m|A^T|v_m\rangle^* = -\langle u_m|A|v_m\rangle$$

Substitute $A|v_m\rangle = -\epsilon_m|u_m\rangle$ into this expression:

$$-\langle u_m|A|v_m\rangle = -\langle u_m|(-\epsilon_m|u_m\rangle) = \epsilon_m\langle u_m|u_m\rangle$$

Equating the two results for $\langle v_m|A|u_m\rangle$:

$$\implies \epsilon_m\langle u_m|u_m\rangle = \epsilon_m\langle v_m|v_m\rangle$$

$$\implies \langle u_m|u_m\rangle = \langle v_m|v_m\rangle = \frac{1}{2} \quad (\text{see Before: } \langle u_m|u_m\rangle + \langle v_m|v_m\rangle = 1)$$

$$\implies \|u_m\|^2 = \|v_m\|^2 = \frac{1}{2}$$

i.e., the normalized real vectors here are $\sqrt{2}|u_m\rangle$ and $\sqrt{2}|v_m\rangle$.

Since we have motivated how $A$ acts like a rotation in $2N$-dim real space, we seek to rotate our (real) basis to arrive at the required block-diag form of $A$. We create a matrix $Q \in SO(2N)$ with the appropriate real vectors as columns:

$$Q = \begin{pmatrix} \sqrt{2}|u_1\rangle & \sqrt{2}|v_1\rangle & \sqrt{2}|u_2\rangle & \dots & \sqrt{2}|u_N\rangle & \sqrt{2}|v_N\rangle \end{pmatrix}$$

We perform the transform $Q^T AQ$, and focus on the $m$-th $2\times 2$ block (Say, $B_m$ be the submatrix):

$$B_m = 2\begin{pmatrix} \langle u_m|A|u_m\rangle & \langle u_m|A|v_m\rangle \\ \langle v_m|A|u_m\rangle & \langle v_m|A|v_m\rangle \end{pmatrix}$$

We already know the action of $A$ on these real vectors:

$$\begin{aligned} \langle u_m|A|u_m\rangle &= 0 \\ \langle v_m|A|v_m\rangle &= 0 \\ \langle v_m|A|u_m\rangle &= \langle v_m|\epsilon_m|v_m\rangle = \epsilon_m\left(\frac{1}{2}\right) = \frac{\epsilon_m}{2} \\ \langle u_m|A|v_m\rangle &= \langle u_m| - \epsilon_m|u_m\rangle = -\epsilon_m\left(\frac{1}{2}\right) = -\frac{\epsilon_m}{2} \end{aligned}$$

So,

$$B_m = 2\begin{pmatrix} 0 & -\epsilon_m/2 \\ \epsilon_m/2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\epsilon_m \\ \epsilon_m & 0 \end{pmatrix}$$

This is skew-symmetric $\rightarrow$ To make it diagonal, all we need to do is reorder the real vector basis, i.e. $(\sqrt{2}|v_m\rangle, \sqrt{2}|u_m\rangle)$ instead of the other way around.

With this in mind, we define normal-mode Majorana operators by combining the $c$-Majorana operators via $Q \in SO(2N)$:

$$f_k = \sum_{j=1}^{2N} c_j Q_{jk}$$

Which is easily invertible:

$$c_j = \sum_{p=1}^{2N} Q_{jp} f_p$$

Verify that $f_k$'s still obey Majorana anticommutation relation $\rightarrow$:

$$\begin{aligned} \{f_k, f_l\} &= \sum_{j=1}^{2N}\sum_{m=1}^{2N} Q_{jk}Q_{ml}\{c_j, c_m\} \\ &= \sum_{j=1}^{2N}\sum_{m=1}^{2N} Q_{jk}Q_{ml}(2\delta_{jm}) \\ &= 2\sum_{j=1}^{2N} Q_{jk}Q_{jl} \\ &= 2(Q^TQ)_{kl} \end{aligned}$$

Since $Q \in SO(2N)$, $Q^TQ = \mathbb{I}$, so $(Q^TQ)_{kl} = \delta_{kl}$. Thus, $\{f_k, f_l\} = 2\delta_{kl}$, which satisfies the condition. So, we can conclude that $f_k$ are valid Majorana operators, and $f_k^\dagger \equiv f_k$, $f_k^2 = 1$.

We put the defining expression for these normal modes into the itinerant $c$-Majorana Hamiltonian, i.e., put $c_j = \sum_{p=1}^{2N} Q_{jp} f_p$ into:

$$H = \frac{i}{4}\sum_{j,k=1}^{2N} A_{jk} c_j c_k$$

We get:

$$\begin{aligned} H &= \frac{i}{4}\sum_{j,k=1}^{2N} A_{jk}\left(\sum_{p=1}^{2N} Q_{jp}f_p\right)\left(\sum_{q=1}^{2N} Q_{kq}f_q\right) \\ &= \frac{i}{4}\sum_{p,q=1}^{2N}\left(\sum_{j,k=1}^{2N} Q_{jp}A_{jk}Q_{kq}\right) f_p f_q \\ \implies H &= \frac{i}{4}\sum_{p,q=1}^{2N} (Q^T A Q)_{pq} f_p f_q \end{aligned}$$

Using the block-diagonal form of $A$ in the appropriately ordered real vector basis, we can write:

$$(Q^T A Q)_{pq} = \begin{cases} \epsilon_m, & \text{for } p = \text{odd}, q = \text{even} \\ -\epsilon_m, & \text{for } p = \text{even}, q = \text{odd} \\ 0, & \text{otherwise} \end{cases}$$

This means:

1. If $p = 2m-1, q = 2m \implies (Q^T A Q)_{pq} = \epsilon_m$
2. If $p = 2m, q = 2m-1 \implies (Q^T A Q)_{pq} = -\epsilon_m$

We may restate the sum in the Hamiltonian (over $p, q$) as a sum over all modes $m \in \{1 \dots N\}$:

$$\begin{aligned} H &= \frac{i}{4}\sum_{m=1}^{N}\left[(Q^T A Q)_{2m-1,2m} f_{2m-1} f_{2m} + (Q^T A Q)_{2m,2m-1} f_{2m} f_{2m-1}\right] \\ &= \frac{i}{4}\sum_{m=1}^{N}\left[\epsilon_m f_{2m-1} f_{2m} - \epsilon_m f_{2m} f_{2m-1}\right] \end{aligned}$$

Using the Majorana anticommutation on the 2nd term ($f_{2m} f_{2m-1} = -f_{2m-1} f_{2m}$):

$$\begin{aligned} \implies H &= \frac{i}{4}\sum_{m=1}^{N}\left[\epsilon_m f_{2m-1} f_{2m} + \epsilon_m f_{2m-1} f_{2m}\right] \\ &= \frac{i}{2}\sum_{m=1}^{N} \epsilon_m f_{2m-1} f_{2m} \end{aligned}$$

As the final step, we re-form complex Dirac fermions from the real Majorana fermions! Why this choice? Let us check if it recovers the standard fermionic anticommutation:

$$\begin{aligned} \{a_m, a_n^\dagger\} &= \frac{1}{4}\{f_{2m-1} + i f_{2m}, f_{2n-1} - i f_{2n}\} \\ &= \frac{1}{4}\Big(\{f_{2m-1}, f_{2n-1}\} - i\{f_{2m-1}, f_{2n}\} + i\{f_{2m}, f_{2n-1}\} + \{f_{2m}, f_{2n}\}\Big) \end{aligned}$$

Now, if $m = n$, only the 1st and 4th terms survive (since cross terms anticommute to 0):

$$\{a_m, a_m^\dagger\} = \frac{1}{4}[2 + 0 + 0 + 2] = 1 \quad (\text{if } m = n)$$

Otherwise, all terms vanish.

$$\implies \{a_m, a_n^\dagger\} = \delta_{m,n}$$

Which is the expected relation.
Note that swapping the odd and even indices in the definition of $a_m$ does not change anything. We have freedom of choice there.

The number operator for the $a$-complex fermions is then:

$$\begin{aligned} a_m^\dagger a_m &= \frac{1}{4}(f_{2m-1} - if_{2m})(f_{2m-1} + if_{2m}) \\ &= \frac{1}{4}(f_{2m-1}^2 + if_{2m-1}f_{2m} - if_{2m}f_{2m-1} + f_{2m}^2) \end{aligned}$$

Using properties of $f$-Majoranas ($f_k^2 = 1$ and $f_j f_k = -f_k f_j$ for $j \neq k$):

$$\begin{aligned} \implies a_m^\dagger a_m &= \frac{1}{4}(1 + 1 + 2if_{2m-1}f_{2m}) \\ &= \frac{1}{2} + \frac{i}{2}f_{2m-1}f_{2m} \end{aligned}$$

So, inverting this:

$$\hookrightarrow \frac{i}{2}f_{2m-1}f_{2m} = a_m^\dagger a_m - \frac{1}{2}$$

Put this into the expression of the Hamiltonian that we last derived:

$$\begin{aligned} H &= \sum_{m=1}^{N} \epsilon_m \left(\frac{i}{2}f_{2m-1}f_{2m}\right) \\ \implies H &= \sum_{m=1}^{N} \epsilon_m \left(a_m^\dagger a_m - \frac{1}{2}\right) \end{aligned}$$

Which is our diagonalized Hamiltonian!

**Note:** The ground state energy is $E_0 = -\frac{1}{2}\sum_{m=1}^{N} \epsilon_m$